Project Planning Phase
Planning Logic

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Project Name	Online payment fraud detection using machine learning

Data Collection

In this phase, we collect historical transaction data required to train the machine learning model. The dataset may include features such as transaction ID, transaction amount, time, location, device details, payment method, and fraud label (fraud/not fraud).

Data can be collected from:

- Public datasets (e.g., credit card fraud datasets)
- Simulated transaction records
- Banking or payment gateway logs (if available)

The collected data should contain both fraudulent and non-fraudulent transactions to help the model learn patterns effectively.

Data Cleaning (Data Pre-processing)

Raw data may contain missing values, duplicate records, irrelevant columns, or inconsistent formats. Cleaning ensures high-quality input for model training.

Steps involved:

- Removing unnecessary columns (e.g., transaction ID if not useful)
- Handling missing values (drop or fill with mean/median)
- Removing duplicate records
- Encoding categorical variables (e.g., payment type, device type)
- Feature scaling (StandardScaler/MinMaxScaler)
- Handling class imbalance using techniques like SMOTE (since fraud cases are usually very low)

This step improves model accuracy and reduces noise in the dataset.

Exploratory Data Analysis (EDA)

EDA helps us understand the dataset and identify patterns related to fraud.

Univariate Analysis

- Distribution of transaction amounts
- Count of fraud vs non-fraud transactions
- Frequency of different payment methods

Bivariate Analysis

- Relationship between transaction amount and fraud
- Location vs fraud occurrence
- Time of transaction vs fraud cases

Descriptive Analysis

- Mean, median, standard deviation
- Correlation matrix to identify important features
- Visualization using bar charts, histograms, heatmaps

EDA helps in identifying important features and understanding fraud behavior patterns before model building.